SAMPLE
This is a real AS Paper 2 I marked for a student, shown as they got it back, in their own handwriting. The only change is that their name is removed. Every mark and note is what they received.

YOUR OWN PAPER
Want yours marked like this? It's from £25 a paper at **cieinsider.com/marked**. The last page explains how it works.

Here's your Paper 2.

9702/23, October/November 2024 (Paper 2, Variant 3). Marked against Cambridge's published mark scheme for that paper, line by line.

47 / 60
YOUR MARK

44 / 60
WORST CASE

Q1	Q2	Q3	Q4	Q5	Q6	Q7
6/7	11/11	7/9	3/7	6/8	10/10	4/8

That's an A on this paper: the A line for 9702/23 in that series was 47 out of 60. Worst case: if your examiner was having a bad day, the three marks I gave you on the benefit of the doubt could go the other way, and 47 becomes 44, a B. You're exactly on the A line, so the green cards on page 3, which make those three marks safe, matter as much as the marks you lost. Your calculations are the strength. Every one of them scored full marks except 4(b)(i), and there the arithmetic was right too. Almost everything you lost was in drawings and in recall.

The one thing I'd fix before your next Paper 2

When a question asks you to sketch or mark something, work the physics out first, then draw it. Three drawing parts cost you 4 marks, on a paper where you could calculate everything:

- **1(b)(i)**: two parts later you worked out that the parcel lands with 35 m/s down and 42 m/s across. That's about 40 degrees below the horizontal, so the path can't end nearly vertical. (note 1)
- **3(c)**: the end point can be calculated. x is zero when the upthrust holds up the whole 4.0 N, at h = 0.29 m. (note 3)
- **5(b)(ii)**: follow the source round the circle before you draw anything. (note 7)

The second thing is quicker: **learn the quark charges**. 7(a) was 3 marks of recall, and every entry was wrong. (note 8) The third is the unit: **quote the unit even when you're copying the value from the question**. That cost 4(b)(ii) a mark. (note 6)

Where the other marks went

The blue numbers match the blue circles on your script.

1 **1(b)(i)** 1 MARK

You did the hard part: your path starts horizontally from the bottom of the aircraft. But there's no air resistance, so the horizontal velocity stays 42 m/s all the way down. The path never turns vertical. Your own answers to (iii) and (iv) give the landing direction: $\tan^{-1}(35/42)$ is about **40 degrees** below the horizontal. Your curve finishes at nearly 80 degrees.

> EXAMINER REPORT, NOV 2024
> "Many candidates gave a parabola with a vertical final section even though the question stated that there was no air resistance."

2 **3(a)** 1 MARK

Leave out "dynamic". The principle of moments applies to any body in equilibrium, and the beam in this question isn't moving. Write: **for a body in equilibrium, the sum of the clockwise moments about a point equals the sum of the anticlockwise moments about the same point.**

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3 **3(c)** 1 MARK

The start point and the straight line with a negative gradient both scored. The end point is nearly a full small square too far to the right: your line reaches x = 0 at about h = 0.295 m, and it needs to finish at 0.29 m. You can calculate it. When x = 0 the load has no moment, so the upthrust carries the whole 4.0 N. The upthrust is proportional to h (1.4 N at 0.10 m), so h = 0.10 × 4.0/1.4 = **0.29 m** (0.286 before rounding). Draw the line to that point on the axis.

4 **4(a)(i)** 1 MARK

Stress is force per unit **cross-sectional** area. Without "cross-sectional" the mark isn't given.

> EXAMINER REPORT, NOV 2024
> "Many candidates were not specific and did not state that the force needed to be applied on the cross-sectional area."

5 **4(b)(i)** 2 MARKS

The load hangs halfway between the two wires, so each wire carries half of it: **9.0 N**, not 18 N. Then A = (9.0 x 0.84)/(1.9 × 10^9 x 0.47 × 10^{-3}) = **8.5 × 10^{-6} m^2**. Your equation and your arithmetic were both right.

> EXAMINER REPORT, NOV 2024
> "Many candidates used the full load for the force causing the extension of wire X, instead of only half the value of the load."

6 **4(b)(ii)** 1 MARK

Naming the force, the length and the extension scored the first mark. The second mark is for the conclusion about wire Y, and it's lost on two points. You never say that Y has the **greater cross-sectional area**, and "less than 1.9 × 10^9" has **no unit**. The unit is part of the answer even when the number is copied from the question: **1.9 × 10^9 Pa**. A sentence that scores both: **F, L and x are the same for both wires. Wire Y has the greater cross-sectional area, and E = FL/Ax, so the Young modulus of Y is less than 1.9 × 10^9 Pa.**

7 **5(b)(ii)** 2 MARKS

Follow the source round once. At t = 0 it's moving across the observer's line of sight, so the observer hears **780 Hz**. Over the top of the circle it moves away, so the frequency **falls to a minimum** (about 695 Hz). At the far side it's moving across again: back to 780 Hz. Along the bottom it moves towards the observer, so it **rises to the maximum** (your 890 Hz). It finishes back at 780 Hz. That's one smooth wave: start in the middle, down, up, back to the middle. A step can't be right, because the speed along the line of sight changes smoothly.

> EXAMINER REPORT, NOV 2024
> "The majority drew graphs that showed a variation for half a rotation only and many were an incorrect shape."

8 **7(a)** 3 MARKS

The charges are **up +2/3, down -1/3, strange -1/3**. Each antiquark has the opposite sign: **anti-up -2/3, anti-down +1/3, anti-strange +1/3**. You swapped the 1/3s and 2/3s, and gave the anti-up the same sign as the up. A quick check: a proton is uud, so +2/3 +2/3 -1/3 = +1.

9 **7(c)** 1 MARK

Up quark to down quark scored. The second mark is for naming both particles emitted: a **positron** and an electron neutrino. You named the neutrino in words, but the positron only appears as a symbol in your equation, and "describe" needs the name. Your equation is right, by the way.

> EXAMINER REPORT, NOV 2024
> "The positron was rarely named as a fundamental particle emitted with a neutrino."

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Marks you got, but I'd tighten the wording

These all scored. Three were marked generously (BOD), and the others are habits worth fixing. Don't give an examiner any excuse to take a mark away. They mark hundreds of scripts and won't puzzle over yours, so make every step visible. You don't need full sentences: step-by-step maths, with a few words to signpost each step, is clearer and quicker.

Show-that questions need this most. The answer is printed in the question, so no mark is ever for the final answer. Every mark is for a step, and a step you skip or leave the examiner to guess is a mark at risk.

A **3(b)(i)**

Full marks, and the same route as the mark scheme. Two things to tighten. F appears from nowhere, so the examiner has to work out what it is: say what it stands for. And the question asks you to show that the upthrust **U** is 1.4 N, so write **"U ="** on the last line. You wrote "4.0 - 2.6 N = 1.4 N", leaving the examiner to decide it's U. One step per line:

> F = tension in the string holding the cylinder
> Moments about the pivot: 2.6 × 0.40 = F x 0.40
> F = 2.6 N
> U = 4.0 - F = 4.0 - 2.6 = 1.4 N

B **5(a)(i)** BOD

Your antinode cross is in exactly the right place along the string, so it scored. But it sits on the dashed line, not on the wave. An antinode is where the string moves furthest, so put the cross **on the peak or the trough**, where the string is at that instant.

C **5(a)(ii)** BOD

This is a show-that, and it says "explain your reasoning", so the one mark is for the reasoning, not the 0.32 m. "3/2 lambda = 0.48" starts at the answer to the reasoning: the examiner has to go back to the diagram to see where the 3/2 came from. I gave it, but another examiner might not. You don't need sentences, just that first step, signposted:

> XY = 3 node-to-node gaps = 3 x lambda/2
> 3 x lambda/2 = 0.48 m
> lambda = 0.48 × 2/3 = 0.32 m

D **6(a)**

"Voltage" scored, but the wording to use is **potential difference per unit current**. Never write a unit ("volt per amp"): a definition is in terms of quantities.

E **6(c)(ii)** BOD

Your conclusion is right, but P is not proportional to I. For the wire, R is fixed, so **$P = I^2R$** and P is proportional to I^2. Write the relationship you actually mean:

> As the temperature rises, the thermistor's resistance decreases, so the total resistance decreases and the current increases. The wire's resistance is fixed, so the power in it, $P = I^2R$, increases.

Reading the marks on your script

- ✓ Mark awarded, placed where you earned it
- ✓ In a calculation, all the ticks go by the final answer
- ✗ Not awarded
- ∧ Something the mark needed is missing here
- 2/2 Your mark for that part
- BOD Marked generously. See the green cards for the safer wording

Good luck with the real one.
Rich

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Cambridge Assessment International Education

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

* 6 3 7 7 6 9 7 3 7 8 *

PHYSICS **9702/23**

Paper 2 AS Level Structured Questions **October/November 2024**

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 60.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.

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Data

acceleration of free fall	g	=	$9.81\ \text{m s}^{-2}$
speed of light in free space	c	=	$3.00 \times 10^{8}\ \text{m s}^{-1}$
elementary charge	e	=	$1.60 \times 10^{-19}\ \text{C}$
unified atomic mass unit	$1\ \text{u}$	=	$1.66 \times 10^{-27}\ \text{kg}$
rest mass of proton	m_p	=	$1.67 \times 10^{-27}\ \text{kg}$
rest mass of electron	m_e	=	$9.11 \times 10^{-31}\ \text{kg}$
Avogadro constant	N_A	=	$6.02 \times 10^{23}\ \text{mol}^{-1}$
molar gas constant	R	=	$8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Boltzmann constant	k	=	$1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	G	=	$6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
permittivity of free space	ε_0	=	$8.85 \times 10^{-12}\ \text{F m}^{-1}$
	$\left(\frac{1}{4\pi\varepsilon_0}\right.$	=	$\left.8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
Planck constant	h	=	$6.63 \times 10^{-34}\ \text{J s}$
Stefan–Boltzmann constant	σ	=	$5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
hydrostatic pressure	Δp	=	$\rho g \Delta h$
upthrust	F	=	$\rho g V$
Doppler effect for sound waves	f_o	=	$\frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 **(a)** Define acceleration.

the rate of change of velocity ✓ 1/1

..... [1]

(b) A small aircraft is flying horizontally at a speed of 42 m s^{-1} at a height of 63 m above horizontal ground, as shown in Fig. 1.1.

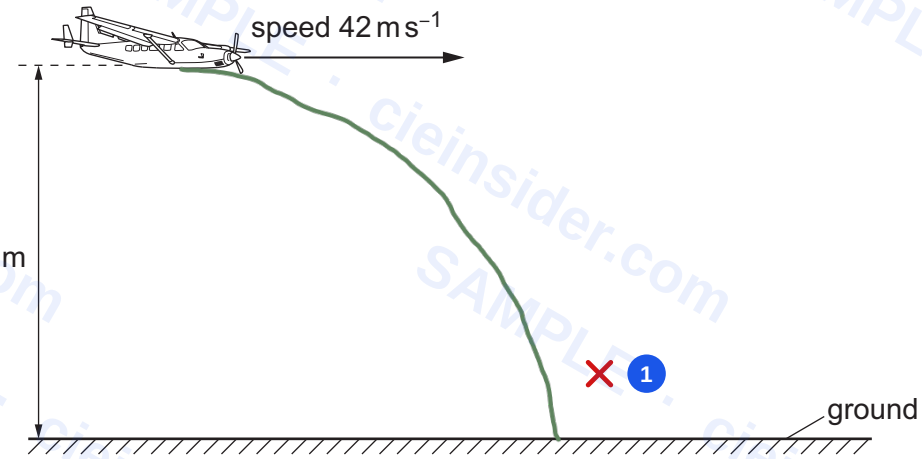


Fig. 1.1

The aircraft drops a small parcel. The parcel is released from the aircraft at the instant shown in Fig. 1.1. Air resistance is negligible.

(i) On Fig. 1.1, draw a line to show the path of the parcel as it falls from the aircraft to the ground. [1] 0/1

(ii) Calculate the time taken from the instant of release to the instant the parcel reaches the ground.

t = 3.6s

$s = ut + \frac{1}{2}at^2$

$-63 = (0)t + \frac{1}{2}(-9.81)t^2$

$t^2 = \frac{63}{4.905}$

$t = \sqrt{\frac{63}{4.905}}$

2/2

time = 3.6 ✓✓ s [2]

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(iii) Calculate the vertical component of the velocity of the parcel immediately before it reaches the ground.

$v^2 = u^2 + 2as$

$v^2 = 0^2 + 2(-9.81)(-63)$

$v^2 = 1236.06$

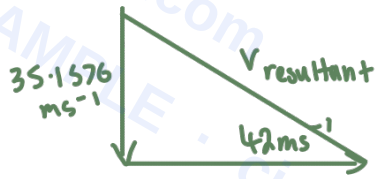
$v = 35.2$

$= 35\ ms^{-1}$

1/1

vertical component of velocity = ... 35 ✓ ..... $m\,s^{-1}$ [1]

(iv) Determine the speed at which the parcel reaches the ground.



$V_{resultant} = \sqrt{35.1576^2 + 42^2}$

$= 54.77$

$= 55\ ms^{-1}$

2/2

speed = ... 55 ✓✓ ..... $m\,s^{-1}$ [2]

[Total: 7]

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2 **(a)** State the principle of conservation of momentum.

total initial momentum is equal to total final momentum in an isolated system. ✓✓

2/2

[2]

(b) A ball X has mass 240 g and moves in a straight line on a horizontal frictionless surface with an initial speed of 16 m s^{-1}. The ball collides with a stationary ball Y that has mass 480 g. After the collision, ball X is stationary, as shown in Fig. 2.1.

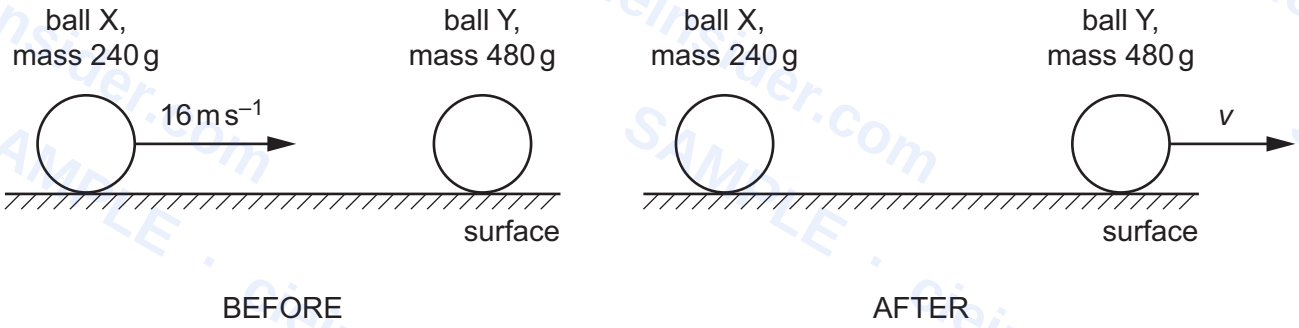


Fig. 2.1

(i) Show that the speed v of ball Y after the collision is 8.0 m s^{-1}.

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$240 \times 16 = 480 \times v_2$

$v_2 = \frac{3840}{480}$

$= 8.0\ ms^{-1}$ ✓

1/1

[1]

(ii) Calculate the change in the total kinetic energy ΔE_K of the balls due to the collision.

initial $= \frac{1}{2}mv^2$

$= \frac{1}{2}(0.24)(16)^2$

$= 30.72$ J

final $= \frac{1}{2}mv^2$

$= \frac{1}{2}(0.48)(8)^2$

$= 15.36$

$\Delta E_K = 30.72 - 15.36$

$= 15.36$

$= 15$ J

3/3

ΔE_K = ... 15 ✓✓✓ ..... J [3]

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(c) The collision in **(b)** lasts for a time of 2.0 ms. Assume that the contact force between the balls is constant during this time.

(i) Determine the magnitude and direction of the force exerted on ball X by ball Y during the collision.

$F = \frac{\Delta p}{\Delta t}$

$= \frac{(0.48 \times (8-0))}{2 \times 10^{-3}}$

$= 1920\ N$

magnitude = ... 1900 ✓✓ ..... N

direction ..... left ✓ .....

3/3

[3]

(ii) Compare the magnitude and direction of the force exerted on ball Y by ball X during the collision with the answers in **(c)(i)**. No further calculations are required.

the magnitude of the force is equal in magnitude ✓

but opposite in direction ✓

2/2

[2]

[Total: 11]

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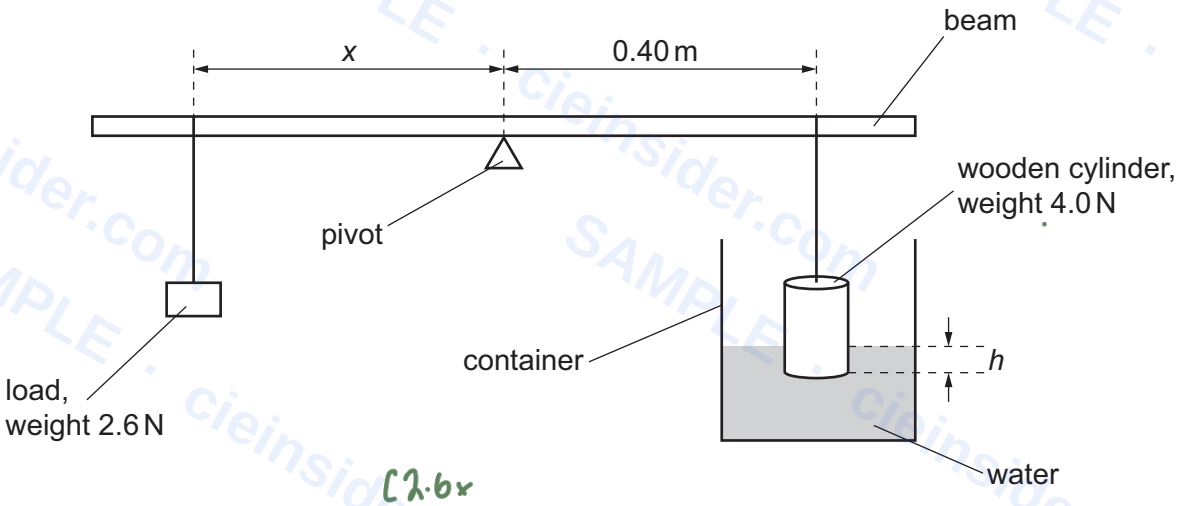
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3 **(a)** State the principle of moments.

total clockwise moment is equal to total anticlockwise moment in a system in dynamic equilibrium

[1] 0/1

(b) A rigid uniform beam rests on a pivot at its centre, as shown in Fig. 3.1.



[2.6×

Fig. 3.1 (not to scale)

A load of weight 2.6 N is suspended from the beam at distance *x* from the pivot.

A wooden cylinder of weight 4.0 N is suspended from the beam at a distance of 0.40 m from the pivot on the opposite side of the pivot to the load. The cylinder rests in a container of water. The lower part of the cylinder is immersed in the water to depth *h*.

Initially, *h* is equal to 0.10 m and *x* is equal to 0.40 m. The system is in equilibrium.

(i) Use the principle of moments to show that the upthrust *U* exerted by the water on the cylinder is 1.4 N.

(2.6 × 0.4) = (0.4 × F)

F = 2.6 N

4.0 − 2.6 N = 1.4 N ✓✓

2/2

[2]

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(ii) The density of the water is $1.0 \times 10^3\,\text{kg m}^{-3}$.

Calculate the area A of the circular cross-section of the cylinder.

$U = \rho g V$

$U = \rho \times g \times (0.1 \times A)$

$1 \times 10^3 \times 9.81 \times 0.1 \times A = 1.4$

$A = 1.4 \times 10^{-3}\,\text{m}^2$ ✓✓✓

3/3

A = 1.4×10^{-3} m^2 [3]

(c) More water is gradually added to the container in **(b)**, so that depth h in Fig. 3.1 gradually increases. The length x is continuously adjusted so that the system remains in equilibrium.

On Fig. 3.2, sketch the variation of x with h. Use the space below for any working.

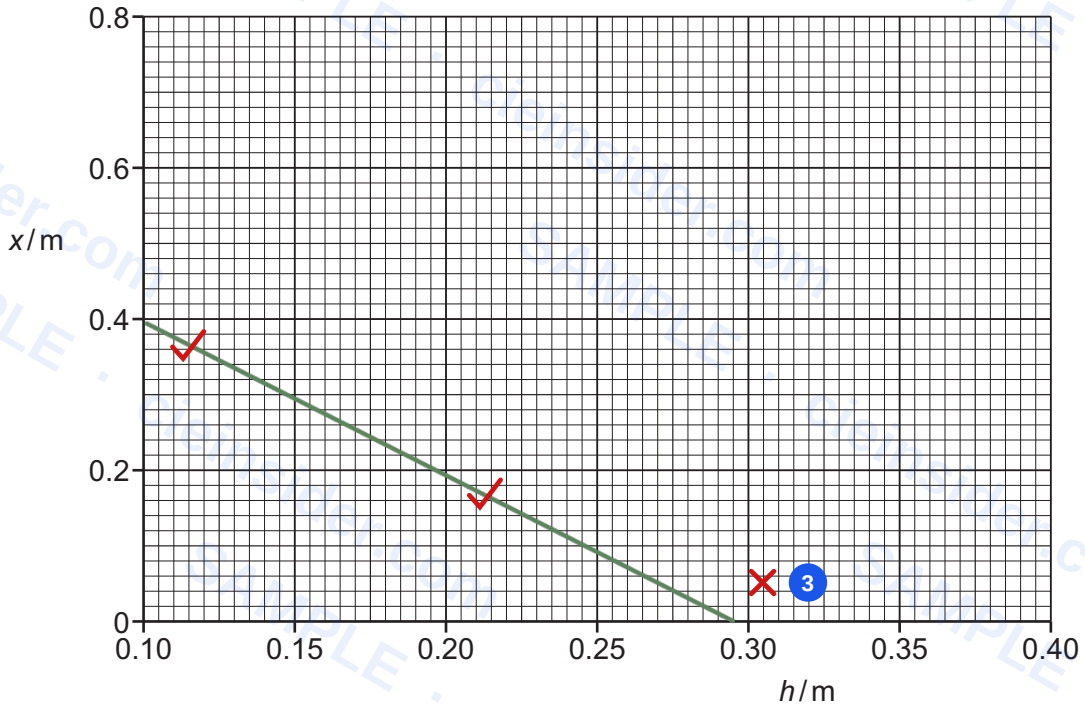


Fig. 3.2

2/3

[3]

[Total: 9]

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4 **(a)** Define:

(i) stress

force per unit area 4

0/1

..... [1]

(ii) strain.

extension over original length ✓

1/1

..... [1]

(b) Two wires X and Y, with equal unstretched lengths of 0.84 m, are suspended from fixed points that are at the same horizontal level. The lower ends of the wires are attached to a beam of negligible mass. The beam is horizontal and in equilibrium, as shown in Fig. 4.1.

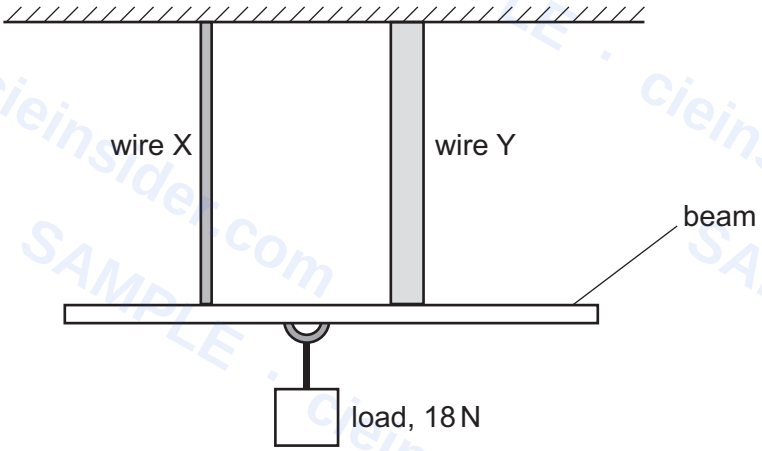


Fig. 4.1

Wire X is made from a metal that has a Young modulus of 1.9×10^9 Pa.
Wire Y is made from a different metal.

A load of weight 18 N is suspended from the beam at a point that is equidistant from the two wires. This load causes both wires to extend by 0.47 mm.

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(i) Determine the cross-sectional area of wire X.

$E = \frac{F \times L}{A \times x}$ ✓

$A = \frac{F \times L}{E \times x}$

$A = \frac{18 \times 0.84}{1.9 \times 10^{9} \times (0.47 \times 10^{-3})}$ ✗ 5

$= 1.7 \times 10^{-5}\ m^2$

1/3

cross-sectional area = ... 1.7×10^{-5} ..... m^2 [3]

(ii) Wire Y has a greater diameter than wire X.

Explain, without calculation, whether the Young modulus of the metal from which wire Y is made is less than, the same as or greater than 1.9×10^{9} Pa.

- Young modulus is less than 1.9×10^{9} ✗ 6
- because $E \propto \frac{1}{A}$, and all other factors (extension, length, applied force) remain the same ✓

1/2

..... [2]

[Total: 7]

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5 **(a)** A stationary wave is formed on a string XY that has a length of 0.48 m. Fig. 5.1 shows the string at one instant in time.

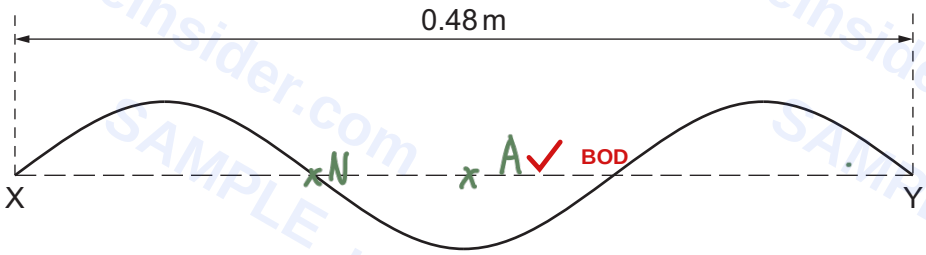


Fig. 5.1

The speed of the wave on the string is 1400 m s^{-1}.

(i) On Fig. 5.1, draw a cross (×) at **one** position that is a node and another cross at **one** position that is an antinode. Label the node N and the antinode A. [1] 1/1

(ii) Show that the wavelength of the wave produced is 0.32 m. Explain your reasoning.

$\frac{3}{2}\lambda = 0.48$

$\lambda = 0.32$ m ✓ BOD

1/1

[1]

(iii) Calculate the frequency of the wave.

$v = f\lambda$

$f = \frac{1400}{0.32}$

$= 4375$

2/2

frequency = ... 4400 ✓✓ ..... Hz [2]

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(b) A source of sound waves of frequency 780 Hz is on a rotating platform. The speed of the source is 39 m s^{-1}.

The sound is detected by an observer that is a large distance from the rotating platform, as shown in Fig. 5.2.

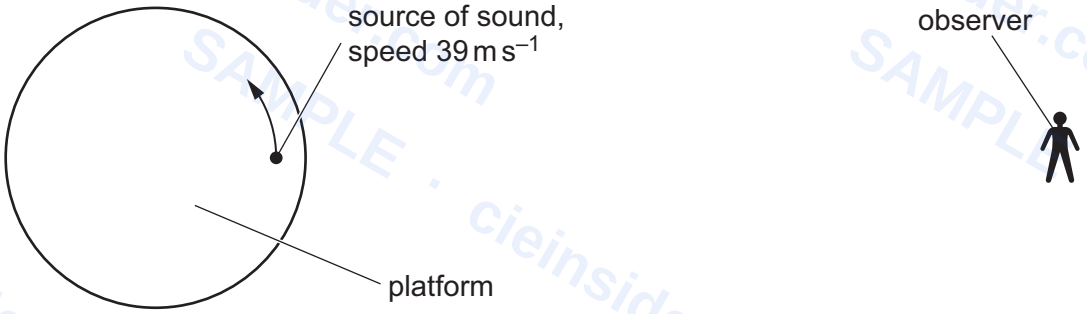


Fig. 5.2 (not to scale)

(i) The speed of sound in air is 320 m s^{-1}.

Calculate the maximum frequency of the sound detected by the observer.

$f_o = f_s \left(\frac{v}{v \pm v_s}\right)$

$= 780 \left(\frac{320}{320 - 39}\right)$

$= 888 \times 2$

$= 890\ Hz$

maximum frequency = ... 890 ✓✓ ..... Hz [2]

2/2

(ii) At time t = 0, the observer detects the sound emitted by the source when it was in the position shown in Fig. 5.2.

On Fig. 5.3, sketch the variation with t of the frequency f of the sound detected by the observer for one complete rotation of the platform. Calculations are not required.

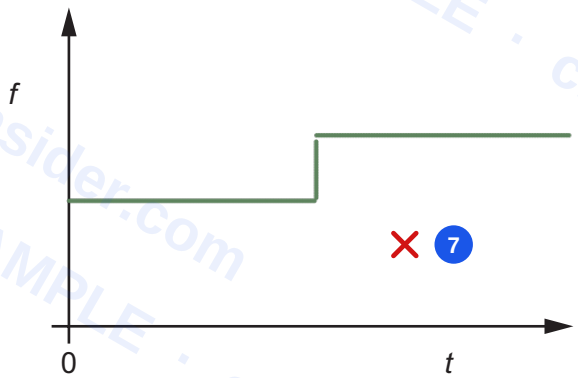


Fig. 5.3

0/2

[2]

[Total: 8]

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6 **(a)** Define resistance.

Voltage per unit current ✓

1/1

..... [1]

(b) A cylindrical metal wire of length 2.4 m and cross-sectional area $8.0 \times 10^{-6}\,m^2$ has a resistance of 0.33 Ω. There is a current in the wire of 4.7 A.

(i) Determine the resistivity of the metal from which the wire is made.

$R = \frac{\rho L}{A}$

$0.33 = \frac{\rho \times 2.4}{8 \times 10^{-6}}$

$\rho = 1.1 \times 10^{-6}\ \Omega m$

2/2

resistivity = 1.1×10⁻⁶ ✓✓ Ω m [2]

(ii) Calculate the charge that passes through the wire in a time of 5.0 minutes.

$Q = I t$

$= 4.7 \times (5 \times 60)$

$= 1410\ C$

$= 1400\ C$

2/2

charge = 1400 ✓✓ C [2]

(iii) The free electrons (charge carriers) in the wire have an average drift speed of $0.16\,mm\,s^{-1}$.

Determine the number density of charge carriers in the metal.

$0.16 mm s^{-1} = 0.16 \times 10^{-3} m s^{-1}$

$I = Anve$

$n = \frac{I}{Ave}$

$= \frac{4.7}{8 \times 10^{-6} \times 0.16 \times 10^{-3} \times 1.6 \times 10^{-19}}$

2/2

number density = 2.3×10^{28} ✓✓ m^{-3} [2]

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(c) The wire in **(b)** may be considered to be a fixed resistor. It is connected in series with a thermistor to a battery that has negligible internal resistance.

(i) Use circuit symbols to complete Fig. 6.1 to show the circuit diagram of this arrangement.



Fig. 6.1

[1] 1/1

(ii) Explain, without calculation, how the power dissipated in the wire changes as the temperature of the thermistor is increased.

• increasing temperature of thermistor decreases its resistance ✓

• current in circuit increases

• $P \propto I$ so, power dissipated in the wire increases ✓ BOD

[2] 2/2

[Total: 10]

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7 **(a)** Complete Table 7.1 to show the charges, in terms of the elementary charge *e*, on each of the flavours of quark and antiquark shown.

Table 7.1

flavour	charge / *e*	
	quark	antiquark
up	$+\frac{1}{3}$ ✗	$+\frac{1}{3}$
down	$-\frac{2}{3}$	$+\frac{2}{3}$ ✗ 8
strange	$-\frac{2}{3}$ ✗	$+\frac{2}{3}$

0/3

[3]

(b) **(i)** State the name of the class (group) of fundamental particles to which baryons and mesons belong.

Hadrons ✓

1/1

..... [1]

(ii) Compare baryons and mesons in terms of their constituent particles.

- Baryons have 3 quarks ✓
- mesons have 1 quark and 1 antiquark ✓

2/2

..... [2]

(c) Describe β^+ decay in terms of the fundamental particles involved.

- a proton turns into a neutron because an up quark changes to a down quark ✓
- and an electron neutrino is also released ^ 9

1/2

..... [2]

$^{210}_{16}X \longrightarrow {}^{210}_{25}Y + {}^{0}_{1}\beta^{+} + \nu$

[Total: 8]

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